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ABSTRACT

Using two-dimensional electronic spectroscopy, we measured the Q_x to Q_y transfer dynamics of the chlorophyll a (Chl a) manifold in the photosystem II (PSII) monomeric core complex from *Arabidopsis thaliana*. A PSII monomeric core consists of 35 Chls a and no Chl b , thus allowing for a clear window to study Chl a Q_x dynamics in a large pigment-protein complex. Initial excitation in the Q_x band results in a transfer to the Q_y band in less than 60 fs. Upon the ultrafast transfer, regardless of the excitation frequency within the Q_x band, the quasi-transient absorption spectra are very similar. This observation indicates that Chl a 's Q_x to Q_y transfer is not frequency selective. Using a simple model, we determined that this is not due to the lifetime broadening of the ultrafast transfer but predominantly due to a lack of correlation between the PSII core complex's Chl a Q_x and Q_y bands. We suggest the origin to be the intrinsic loss of correlation during the Q_x to Q_y internal conversion as observed in previous studies of molecular Chl a dissolved in solvents.

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INTRODUCTION

Photosystem II (PSII) is an important enzyme in the biosphere with the ability to perform the thermodynamically challenging reaction, converting H_2O to molecular oxygen using harvested excitation from sunlight.¹ In higher plants, the supercomplex of PSII usually exists in a dimeric form with the core complex (CC) at the center, surrounded by several peripheral antenna complexes that harvest sunlight and transfer the excitation energy into the reaction center (RC) to perform the photoredox reaction. Understanding the ultrafast excitation energy transfer (EET) dynamics within PSII is, therefore, important to get further insights about the overall photosynthetic mechanism. The PSII CC is a multi-subunit pigment-protein complex, which comprised of a RC, where the water-splitting reaction happens, and two peripheral antennae CP43 and CP47, as shown in Fig. 1(a). The structure of the PSII CC [Fig. 1(a)] is relatively well-conserved among a vast variety of organisms from cyanobacteria, algae, and higher plants.^{1–3}

In green algae and higher plants, the main light-harvesting pigments of PSII are chlorophylls (Chls) a and b . As the main light-harvester molecules, Chls have been extensively studied during the past century. In solvated environments, free Chl molecules exhibit two major absorption bands in the red and blue regions [see Fig. 1(b)], which are named the Q and the Soret (also known as the B) bands, respectively.^{2,5} The Q band is usually comprised of a strong absorption peak, known as the Q_y transition due to the orientation of the corresponding transition dipole moment, and several weaker absorption peaks at shorter wavelengths. The shoulder absorbances in the higher-energy region of the Q band include the Q_x transition and several vibrational progressions of both Q_x and Q_y bands.^{5–8} In general, the Q_x transition and vibrational progressions of Q_y are strongly mixed due to the energy degeneracy and the high density of states.^{3,6,9,10} The degree of mixing between Q_x character and Q_y vibrational procession can be determined via magnetic circular dichroism⁶ or double-crossed polarization two-dimensional spectroscopy.¹⁰ However, within the scope of this study, for

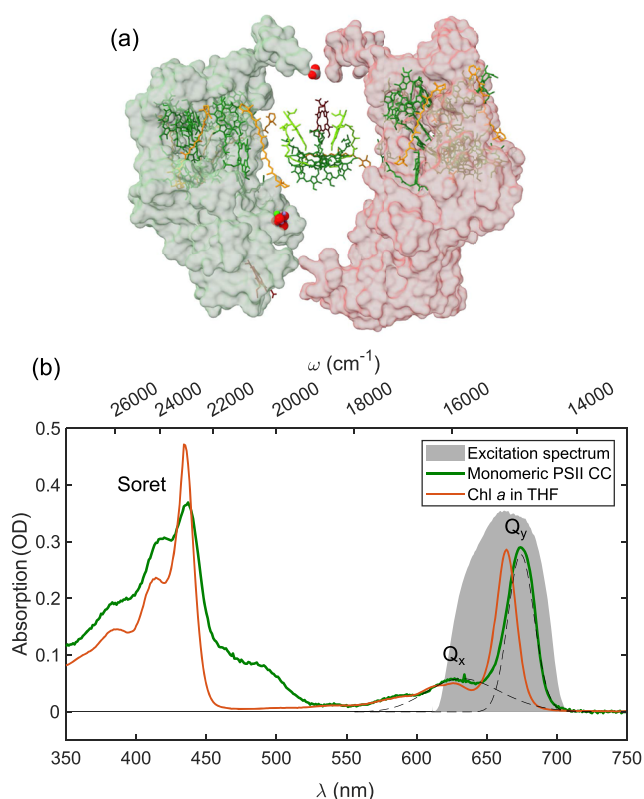


FIG. 1. (a) Structure of the monomeric PSII CC obtained from cryo-electron microscopy (PDB: 7EDA).⁴ The protein scaffold of the antennas CP43 and CP47 is shown in green and red surfaces, respectively. All pigments are shown in sticks with green, light green, orange, and red colors for Chls *a*, Pheo *a*, carotenoids, and heme/Fe-protoporphyrin IX molecules, respectively. The Mn₄CaO₅ cluster and bicarbonate ion are shown in spheres. (b) Linear absorption spectra of the monomeric PSII CC (green) and molecular Chl *a* dissolved in tetrahydrofuran (THF) (orange) overlaid with the excitation spectrum (gray shaded area) used in our 2DES experiments. The black dashed lines are results from Gaussian decomposition to estimate the Q_x and Q_y absorption linewidths of PSII CC.

simplicity, we shall still term the peak that arises as a mixed state between Q_x and the vibronic progressions of Q_y as the Q_x band.

In PSII and other photosynthetic pigment–protein complexes, the Chls are arranged by the protein scaffold and the absorption features, especially the Q_y peak, and pigment orientation are tuned to optimize the EET dynamics happening at femtosecond (fs) to picosecond (ps) timescales.^{11,12} The effects of excitonic couplings and protein binding site modulations can be seen from the Q_y absorption band of PSII CC in Fig. 1(b), which shows a noticeable broadening and redshifts as compared to the Q_y band of molecular Chl *a* dissolved in THF. The resultant multi-excitonic energy levels arisen from excitonic couplings between Chls *a* function as energy funnels to direct the energy to the RC. We note that although within the photosynthetic pigment–protein complexes, such as PSII CC, the Chl *a* pigments exist, both uncoupled and coupled into excitons, for simplicity, we will use the term “molecule” to describe Chls *a* in the PSII CC.

With advanced optical spectroscopic techniques, such as two-dimensional electronic spectroscopy (2DES), various studies have been conducted and rich information about EET, as well as charge separation dynamics in PSII CC and antennas, have been retrieved from the Q_y spectral region. The major trimeric light-harvesting antenna of PSII, light-harvesting complex II (LHCII), was extensively studied with transient absorption (TA),^{13,14} time-resolved fluorescence,¹⁵ and later 2DES,^{16–19} which revealed a highly connected EET network with several Chls *b* → Chls *a* energy transfers spanning from sub-100 fs to several timescale and some slow (~5 ps) energy equilibration dynamics within Chls *a* manifold. The minor antennas of PSII—CP24, CP26,²⁰ and CP29^{21,22}—were also studied using multicolor TA spectroscopy. Slower EET dynamics were observed in CP26 and CP24 compared to LHCII due to the weaker couplings between the Chl pigments.²⁰ Some Chls in CP24 were indicated to be crucial in quenching modulation.²⁰ CP29 was rather more robust with similar dynamics compared to LHCII, and the overall EET efficiency was not affected even by several knock-out mutations.²² A multicolor TA spectroscopic study revealed two different charge separation pathways in PSII RC of green plants at 77 K with the preference determined by different protein conformations.²³ The energy equilibration within the Q_y manifold of PSII RC²⁴ and CC²⁵ was determined with several timescales spanning from ~200 fs to 200 ps at cryogenic conditions via 2DES. More recently, 2DES studies revealed several long-lived (up to 1 ps) quantum coherences in PSII RC,^{26,27} which may assist charge-separation dynamics happening at a ps timescale.^{23,28}

The Q_x band of Chl *a* contributes to a minor but not negligible spectral window where sunlight can be absorbed. This contribution is estimated to be about 20% relative to the Q_y absorption.⁶ Hence, for a more comprehensive understanding of the EET dynamics of such pigment–protein complexes, we will also need to investigate the dynamic behaviors of the energy initially exciting the Q_x band. This information about the Q_x band dynamics is still currently minimal. A time-resolved fluorescence-depletion study determined that the internal conversion (Q_x → Q_y) timescale of molecular Chl *a* dispersed in various organic solvents ranges between 100 and 226 fs.²⁹ The results were later reaffirmed by a more advanced 2DES study³⁰ and high-level quantum–classical molecular dynamics simulations^{31,32} with the determined timescales of ~170 and ~100 to 200 fs, respectively. Another 2DES study measured the loss of correlation when the Q_x excited population relaxed into the Q_y band via center line slope analysis.³³

The dynamics of the Q_x band in pigment–protein complexes, such as PSII, are even more sparse. In this study, we perform the 2DES experiment on the monomeric PSII CC extracted from the green plant *Arabidopsis thaliana*.³⁴ We discuss the ultrafast EET processes within the PSII CC with an emphasis on the EET dynamics of the Q_x band of Chls *a* in the PSII CC, in comparison to the dynamics of the Q_y band. Within the core complex, we can treat Chl *a* as the only functioning pigment as the light-harvester {Pheophytin *a* (Pheo *a*), the de-metallated version of Chl *a* that serves as the primary charge carrier for the electron transfer reaction [see Fig. 1(a)], has minimal contribution to the overall absorption}. Hence, the PSII CC system allows us to study the EET processes in Chl *a*'s Q_x bands without the complexity of the overlaying Chl *b*'s Q_y band, which is present, for example, in LHCII and would complicate the analysis.

In the PSII CC system at physiological temperature, Chl *a*'s Q_y manifold demonstrates complex EET dynamics and only reaches equilibration by 10 ps. This means that for the Q_y band, the dynamics are, in general, dependent on the excitation, which gradually equilibrates and reaches an excitation-independent state in tens of ps. For the Q_x band, it is expected that upon excitation, it will undergo internal conversion and relax to the Q_y band in a timescale of sub-100 fs. This study will then attempt to answer the following questions: during this fast internal conversion as the excitation from Q_x band relaxes into the Q_y manifold, is there a trivial mapping of the Q_x frequencies to the Q_y frequencies and how frequency-selective is this mapping? If the answer to this question is “yes” and “very selective,” respectively, then we can expect that the EET dynamics of the Q_x band will map the dynamics of the Q_y band, i.e., exciting Q_x will be akin to exciting the Q_y band. Using 2DES spectral data, we will, however, show in this paper that the answers are, in fact, negative. We will subsequently estimate the extent of the loss of frequency selectivity and discuss and provide possible reasons behind this phenomenon via a simplified numerical model and references to previous studies in molecular Chl *a*.

MATERIALS AND METHOD

Monomeric PSII CC was prepared according to the previously reported procedure.³⁴ To improve purity, the samples were purified twice on a sucrose gradient. All samples were kept at -80°C until used. For 2DES measurements, the frozen sample was thawed to room temperature and diluted with N_2 -bubbled buffer containing 10 mM HEPES-KOH and 0.01% *n*-dodecyl- α -D-maltopyranoside (α -DDM) until the optical density of the Q_y absorption peak reaches 0.3 mm^{-1} . The sample was loaded into a 1-mm path length quartz cuvette (Starna) and quickly sealed with a cuvette cap and parafilm. The cuvette was rigorously cleaned and purged with N_2 before loading the sample. Linear absorption spectra of the sample were checked before and after every 2DES measurement, and there was no degradation observed.

Details about our experimental setup for 2DES measurement were presented in previous studies.³⁵ Briefly, 700- μJ 1-kHz near-IR pulses coming from a regenerative Ti:sapphire amplifier (Legend Elite, Coherence) were focused into a 2.2-bars Ar-pressurized hollow tube to generate a supercontinuum. After passing through a 750-nm-cutoff shortpass filter, the supercontinuum pulse was split by a wedged fused silica window. The transmitted and reflected beams were used as the pump and the probe beam, respectively. The pump beam was negatively chirped by 21 reflection roundtrips on a chirped mirror pair and then compressed and shaped by a pulse shaper (Dazzler, Fastlite) into a phase-locked double-pulse train having a controllable delay time τ , which was scanned from 0 to 120 fs with 6-fs steps during 2DES measurements. The excitation bandwidth after the pulse shaper was $\sim 1600\text{ cm}^{-1}$ full-width at half-maximum (FWHM) covering the whole Q_y band and a significant portion of Q_x band [see Fig. 1(b)] of the PSII CC absorption spectrum. The probe beam was compressed by a similar chirped mirror pair and delayed by a retroreflector mounted on a motorized linear stage (Physik Instrumente). The polarization plane of the probe pulse is set to be “magic angle” (54.7°) relative to the pump pulse's polarization. The pump, probe, and reference (split from the probe) beams were simultaneously focused into the sample cuvette with a 90° off-axis

parabolic mirror. The transmitted probe and reference beams were then frequency-dispersed on a CCD camera (PIXIS) to resolve the τ -dependent transient absorption (TA) spectra $\Delta A(\tau, T_w, \lambda_t)$, where T_w is the pump-probe delay time. The 2D spectra $S^{2D}(\lambda_\tau, T_w, \lambda_t)$ were obtained after performing numerical Fourier transform along τ dimension. The experimental time resolution was determined to be ~ 60 fs from the temporal cross correlation between the pump and probe pulses measured on neat methanol.

RESULTS

Spectral analysis

Figure 1(b) shows the linear absorption spectrum of PSII CC at room temperature (green line). The main peak centered at 675 nm represents the Q_y band. Another peak centered at 628 nm can also be discerned and represents the Q_x absorption band. The excitation pulse (gray shaded area) covers both the Q_y band and a significant portion of the Q_x transition. The linear spectrum of molecular Chl *a* dissolved in THF is also shown for reference. The absorption linewidths of the PSII CC Q_x and Q_y bands are estimated to be 776 and 240 cm^{-1} half-width at half-maximum (HWHM) via fitting to Gaussian functions [dashed lines in Fig. 1(b)], respectively.

Figures 2(a) and 2(b) show representative 2D spectra of the monomeric PSII CC at room temperature at the waiting times $T_w = 60$ fs and 10 ps, respectively. Each 2D spectrum is a two-dimensional correlation map, which plots the conditional population after a waiting time T_w detected at wavelength λ_t (horizontal axis) upon the excitation at λ_τ (vertical axis). The diagonal signals represent populations persisting at their original excitation frequency, while the off-diagonal signals, i.e., crosspeaks, usually denote population flows from the excitonic states absorbed at λ_τ to the ones probed at λ_t after a delay time T_w . In PSII CC, the different local environments of binding pockets, as well as the excitonic couplings between different Chls *a*, can vary the absorption frequencies of the pigments in both Q_x and Q_y bands. The evolution of 2D peaks with T_w can provide information about the coupling and EET processes happening between these excitonic energy levels. As can be seen from Fig. 1(b), the linear spectrum of monomeric PSII CC exhibits a main absorption of Chls *a*'s Q_y band at 675 nm, which is significantly broadened and red-shifted compared to molecular Chl *a* spectrum due to excitonic coupling and different binding sites in the protein scaffold. In the 2D spectrum at early T_w of 60 fs [Fig. 2(a)], the Q_y signals are clearly differentiated as various diagonal features spanning between $\lambda_\tau = 650\text{--}690$ nm show the energy levels due to excitonic couplings between different Chls. At the later time of 10 ps, the excitation energy has equilibrated within the lowest-energy states and the 2D spectra gradually become λ_τ -independent [see Fig. 2(b)].

We now look at the excitation of the Q_x states in the higher-energy excitation region of $\lambda_\tau = 620\text{--}645$ nm. This spectral region is usually neglected due to the lower signal-to-noise ratios and the very fast relaxation dynamics, i.e., most of the populations excited in the Q_x band relax very rapidly and become Q_y population within 100 fs. In Fig. 2(a), a strong crosspeak at $(\lambda_\tau, \lambda_t) = (630, 685)$ nm appears at $T_w = 60$ fs with no corresponding diagonal peak. This indicates that the $Q_x \rightarrow Q_y$ relaxation was virtually over by 60 fs. Although the timescale of the $Q_x \rightarrow Q_y$ relaxation process cannot be determined precisely due to the limited time-resolution of ~ 60 fs, based on the

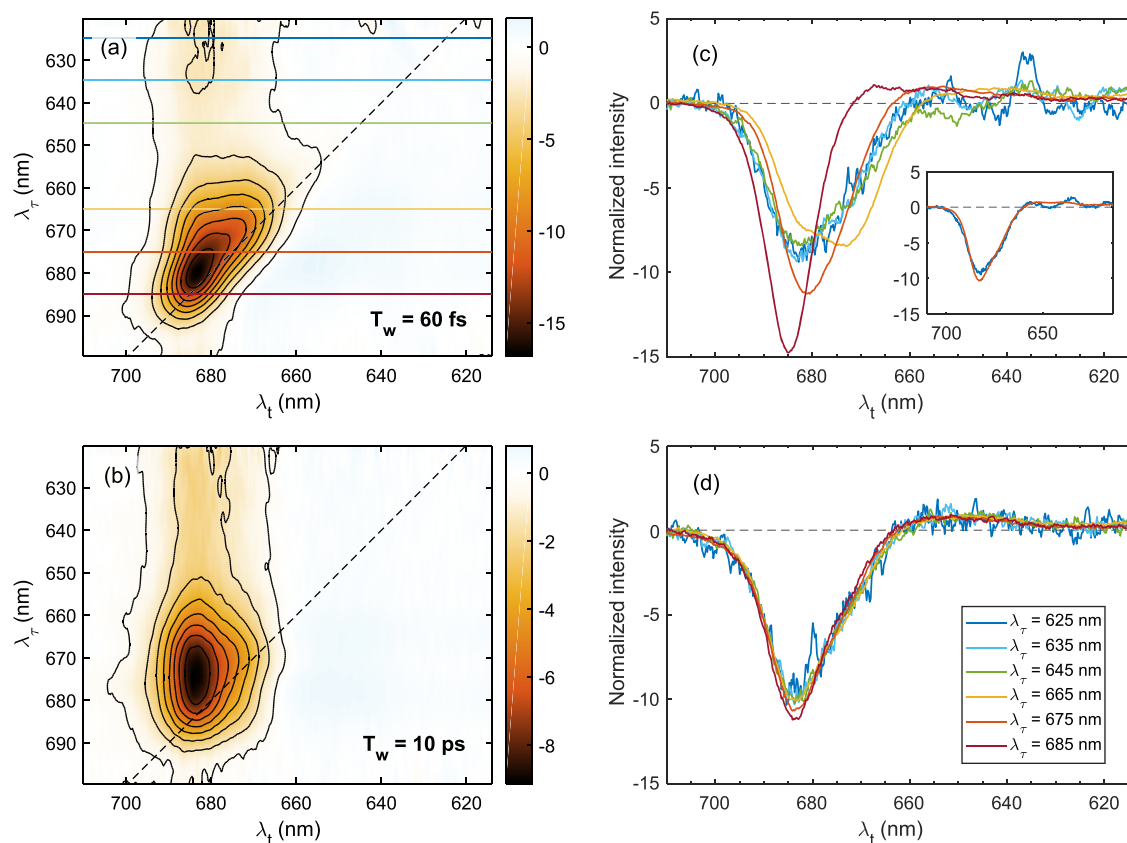


FIG. 2. Representative 2D spectra taken at room temperature of the monomeric PSII CC at $T_w = 60$ fs (a) and 10 ps (b). All presented 2D spectra have been corrected for the excitation spectral dependence.³⁶ Horizontal slices taken from the 2D spectrum at the denoted excitation wavelengths [lines portrayed in (a)] at $T_w = 60$ fs (c) and 10 ps (d). Each slice is averaged within a 2-nm λ_τ bandwidth and area-normalized. The inset of (c) shows the slices averaging across the whole absorption region of the Q_x band (blue, $\lambda_\tau = 625$ – 645 nm) and Q_y band (orange, $\lambda_\tau = 655$ – 695 nm) at 60 fs.

lack of diagonal Q_x signals in the 2D spectra, we can estimate an upper bound of the internal conversion $Q_x \rightarrow Q_y$ time to be ~ 20 fs since the Q_x population is already depleted by 60 fs.

To facilitate the comparison between the excitations at the Q_x and Q_y bands, Figs. 2(c) and 2(d) depict the horizontal slices on the 2D spectra at various excitation wavelengths over 20-nm bandwidth for the Q_x ($\lambda_\tau = 625, 635$, and 645 nm) and the Q_y region ($\lambda_\tau = 665, 675$, and 685 nm) at $T_w = 60$ fs and 10 ps. The cuts are along the lines drawn in Fig. 2(a). Each slice can be interpreted as a quasi-TA spectrum obtained with a narrow-band excitation. In Fig. 2(c), the early time (60 fs) quasi-TA spectra taken at the Q_y region exhibit a strong λ_τ -dependence compared to the ones taken at the Q_x region. The high selectivity of the excited species with respect to the excitation wavelength λ_τ in the Q_y region is consistent with earlier studies^{18,23,24,37} and speaks to the complex nature of the EET networks in the photosynthetic pigment–protein complex. The quasi-TA spectra in the Q_x region at the same T_w have lower signal-to-noise quality due to the smaller signal amplitude originating from the lower transition dipole moment of the Q_x band [cf. Fig. 1(b)]. However, it is clear that the three slices are very similar regardless of the excitation wavelengths. If the Q_x band's frequencies are mapped to the Q_y 's selectively in a straightforward fashion

where the higher/lower frequency end of the Q_x band maps to the higher/lower frequency side of the Q_y band, we will expect to see that the $\lambda_\tau = 625, 635$, and 645 nm slices resemble and follow the trend with the $\lambda_\tau = 665, 675$, and 685 nm slices. The observed lack of resemblance indicates that the $Q_x \rightarrow Q_y$ relaxation is not frequency selective.

The Q_x slices, i.e., $\lambda_\tau = 625, 635$, and 645 nm, are also much broader than the three slices from Q_y ($\lambda_\tau = 665, 675$, and 685 nm) excitation. In fact, they contain signals spanning the whole Q_y band along the λ_t detection axis. This suggests that the quasi-TA spectra excited in the Q_x regions are the combination of all spectra in the Q_y band. This observation is made even clearer by looking at the averaged 2D slices in the whole Q_x and Q_y region at 60 fs plotted in the inset of Fig. 2(c), which shows a significant overlap. Figure 2(d) shows that when the slices evolved to 10 ps, the quasi-TA spectra excited at the Q_x region remain roughly the same, but the ones taken at the Q_y region become very similar to the Q_x spectra. This almost complete overlap indicates that the EET dynamics are virtually completed at 10 ps, which is consistent with the λ_τ -independent 2D spectrum at 10 ps [Fig. 2(b)].

There are two possible major factors that can contribute to the low frequency-selectivity between the Q_x and Q_y bands. One

possible factor is the broad homogeneous linewidth due to the fast relaxation of the Q_x band. Effectively, a fast relaxation rate can broaden the homogeneous linewidth,³⁸ thus increasing the overlap between the Q_x transitions of different chromophores. The other possible factor is that Q_x and Q_y transitions are not correlated, resulting in an inhomogeneous broadening due to a spread of the energy differences between the Q_x and Q_y transitions. In the Simulation section, we will use a simple model to investigate the contribution of these two main factors to the λ_t -independent quasi-TA spectra of the Q_x band excitation.

Simulation

To provide insights from the observations as described above, we perform a numerical simulation to qualitatively reproduce the observed features of quasi-TA spectra. The PSII CC system is modeled as an ensemble of Chls *a*. Each Chl *a* molecule exhibits two transitions representing the Q_x and Q_y bands with the transition frequencies slightly varied to mimic the inhomogeneity of the system. All simulated quasi-TA spectra are the ensemble average of the population-weighted homogeneous line shapes $\mathcal{L}_x(\omega)$ and $\mathcal{L}_y(\omega)$ of the Q_x and the Q_y bands, respectively. The homogeneous line shapes are described as

$$\mathcal{L}_{x/y}^i(\omega) = \frac{\Gamma_{x/y}}{(\omega - \omega_{x/y}^i)^2 + \Gamma_{x/y}^2},$$

where $\Gamma_x = \frac{1}{2\pi T_2^*} + \frac{1}{4\pi\tau_{x \rightarrow y}}$ and $\Gamma_y = \frac{1}{2\pi T_2^*}$ are the HWHM of the Q_x and the Q_y line shapes,³⁸ respectively. The pure-dephasing time $T_2^* = 400$ fs is estimated from the temperature-dependent hole-burning spectra of PSII RC³⁹ and $\tau_{x \rightarrow y}$ is the relaxation time-constant from the Q_x to the Q_y band. The transition frequencies $\omega_{x/y}^i$ of the Q_x and the Q_y bands of the i th chromophore are randomized following Gaussian distributions centering at $\omega_x^0 = 15950$ cm⁻¹ and $\omega_y^0 = 14840$ cm⁻¹. The centers are chosen to replicate the experimental linear absorption spectrum. The correlation between Q_x and Q_y transitions is also controlled during the randomization and is quantified by the Pearson correlation parameter as⁴⁰

$$\rho = \frac{1}{N-1} \sum_{i=1}^N \left(\frac{\omega_x^i - \omega_x^0}{\sigma_x} \right) \left(\frac{\omega_y^i - \omega_y^0}{\sigma_y} \right),$$

where N is the total size of the ensemble and $\sigma_{x/y}$ is the standard deviation of the frequency distribution of the Q_x and Q_y bands, respectively. In this simulation, we use $N = 10\,000$, which is a large enough sample size for a good convergence.

The quasi-TA spectra of the system at T_w after being excited by a narrow-band excitation centered at ω_{exc} are simulated as the ensemble average of the population-weighted homogeneous line shapes as

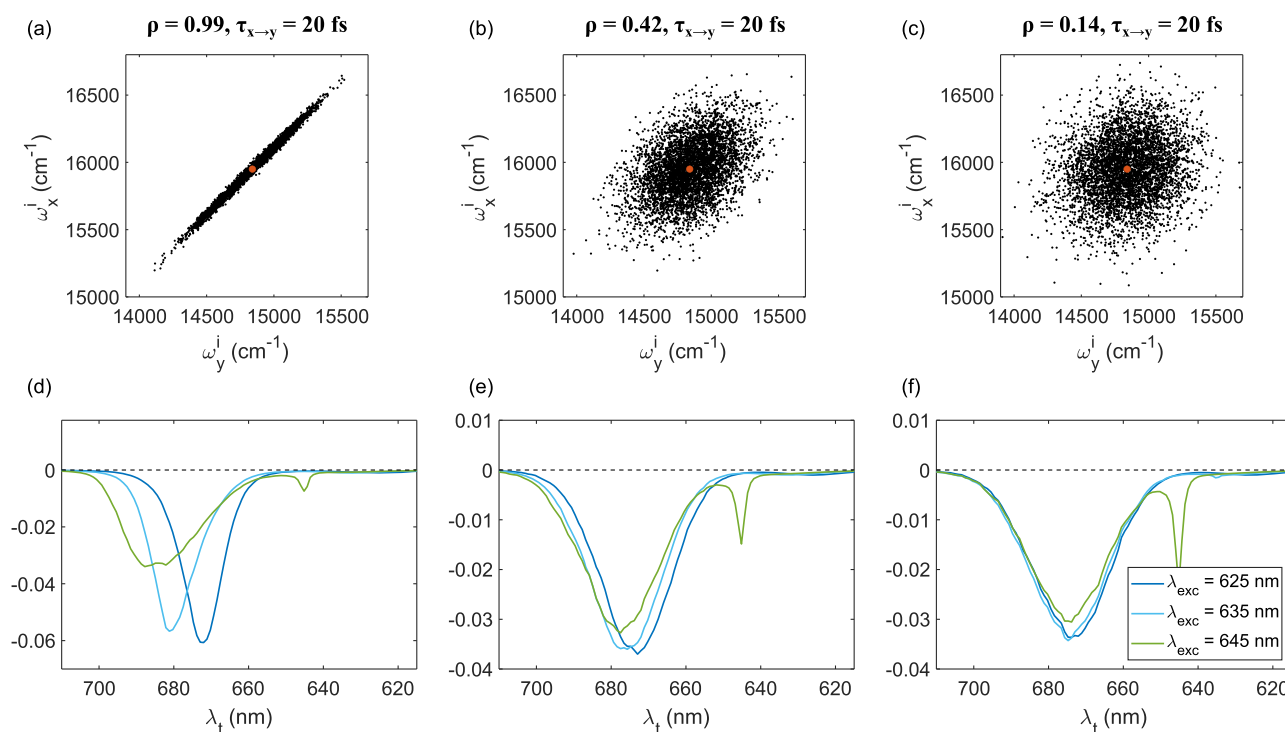


FIG. 3. The effect of correlation parameter ρ on the simulated quasi-TA spectra excited at the Q_x band. The scatter plot of randomized Q_x and Q_y transition frequencies following the correlations $\rho = 0.99$ (a), 0.42 (b), and 0.14 (c). (d)–(f) The corresponding area-normalized quasi-TA spectra of the systems are shown in (a)–(c) at the waiting time $T_w = 60$ fs and the excitation wavelengths $\lambda_{exc} = 625, 635$, and 645 nm.

$$S(\omega_{exc}; T_w, \omega) = - \sum_{i=1}^N \left(\mathcal{L}_x^i(\omega_{exc}) \mu_x^4 e^{-\frac{T_w}{\tau_{x \rightarrow y}}} \mathcal{L}_x^i(\omega) \right. \\ \left. + \left[\mathcal{L}_x^i(\omega_{exc}) \mu_x^2 \mu_y^2 \left(1 - e^{-\frac{T_w}{\tau_{x \rightarrow y}}} \right) \right. \right. \\ \left. \left. + \mathcal{L}_y^i(\omega_{exc}) \mu_y^4 \right] \mathcal{L}_y^i(\omega) \right),$$

where the first and the second terms are the time-dependent population weighted of $\mathcal{L}_x^i(\omega)$ and $\mathcal{L}_y^i(\omega)$ and the negative sign is the conventional indication of the reduced absorption induced by the excitation pulse. μ_x and μ_y are the transition dipole moments associated with the Q_x and Q_y transitions, respectively, which are chosen to replicate the relative intensities of the two absorption bands in the linear spectrum. All simulations are performed in the frequency scale of cm^{-1} and then converted into nm for presentation.

As mentioned in the Spectral analysis section, the diagonal signal of the Q_x band had completely decayed by waiting time $T_w = 60$ fs. To model the depletion of the Q_x population at 60 fs, the slowest relaxation time-constant $\tau_{x \rightarrow y}$ that can give this observation is ~ 20 fs, corresponding to a homogeneous linewidth of $\sim 145 \text{ cm}^{-1}$ (HWHM) of the Q_x band. The simulation is therefore conducted with a constant value of $\tau_{x \rightarrow y} = 20$ fs and varied degree of correlation ρ between the Q_x and Q_y transition energies to investigate the correlation effect on the quasi-TA spectra. The obtained results are presented in Fig. 3. We note that the correlation ρ used here has a general meaning and can also contain contributions from intramolecular spectral diffusion dynamics within the Q_x or Q_y

bands, in addition to the measure of correlation between the Q_x and Q_y transitions. We will consider the effects of intramolecular spectral diffusion in the Discussion section.

Figure 3 presents the simulated quasi-TA spectra excited at the Q_x region at $T_w = 60$ fs with various correlation parameter ρ between Q_x and Q_y transition energies. Despite a broad homogeneous bandwidth of the Q_x band ($\sim 145 \text{ cm}^{-1}$ HWHM), in the extreme case of highly correlated Q_x and Q_y transition energies [Fig. 3(a), $\rho = 0.99$], the simulated quasi-TA spectra exhibit a strong λ_{exc} -dependence, as shown in Fig. 3(d).

The λ_{exc} -dependence of the simulated quasi-TA spectra gradually reduces with lower values of ρ [see Figs. 3(e) and 3(f)]. At a relatively low correlation regime ($\rho = 0.42$), the simulated quasi-TA spectra still vary slightly with different excitation wavelengths [see Fig. 3(e)]. When the correlation is further reduced to $\rho < 0.15$, the λ_{exc} -independent quasi-TA spectra that were experimentally observed in Fig. 2(c) can be reproduced. Clearly, this simulation shows that for $\tau_{x \rightarrow y} \geq 20$ fs, the resultant homogeneous broadening is not sufficient to account for the experimental results and we need to also consider the large non-correlation of the Q_x and Q_y bands.

Note that in the simulated quasi-TA spectra, there is always a sharp feature at $\lambda_t = \lambda_{exc}$, especially for the $\lambda_{exc} = 645 \text{ nm}$. These are the diagonal Q_y bleach signals due to the excitation of the very blue edge of the Q_y band. As they do not result from excitation of the Q_x band, these diagonal Q_y bleach signals do not feature in our analysis.

We now study the effect of the broadened linewidth of the Q_x band due to the ultrafast internal conversion $\tau_{x \rightarrow y} < 20$ fs, i.e., a fast enough relaxation leading to a very broad Γ_x . To investigate

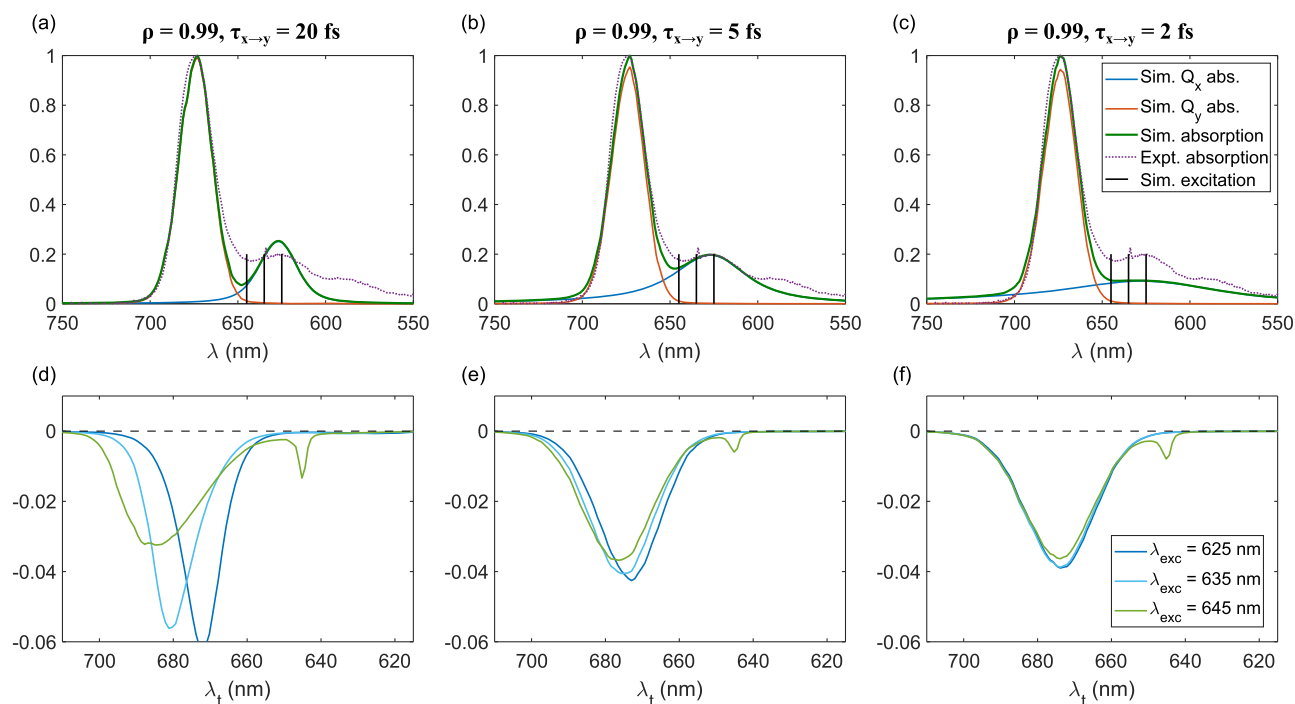


FIG. 4. The effect of internal conversion timescale $\tau_{x \rightarrow y}$ on the simulated quasi-TA spectra excited at the Q_x band. All spectra are simulated with correlation parameter $\rho = 0.99$. Simulated linear absorption spectra with varied $\tau_{x \rightarrow y} = 20$ (a), 5 (b), and 2 fs (c) overlaid with the experimental linear absorption spectra of PSII CC at room temperature. The corresponding quasi-TA spectra are presented similarly as in Figs. 3(d)–3(f).

the effect of line shape broadening, the correlation parameter ρ is kept as 0.99 to restrict the system in the high correlation regime. The quasi-TA spectra are simulated with various $\tau_{x \rightarrow y}$, and the results are presented in Fig. 4. The black lines in Figs. 4(a)–4(c) indicate the excitation wavelengths λ_{exc} used to simulate the quasi-TA spectra.

Figures 4(a) and 4(d) basically present the same data as in Figs. 3(a) and 3(d) with highly correlated Q_x and Q_y transitions and the relaxation time-constant $\tau_{x \rightarrow y} = 20$ fs. As described above, the simulated quasi-TA spectra in this case still exhibit very strong λ_{exc} -dependence, in contrast to the experimental observations (see Fig. 2). To retrieve the similar margin of λ_{exc} -independence as observed in the experimental spectra, the internal conversion time has to be reduced to 2 fs, yielding an unrealistically broad linewidth of 1340 cm^{-1} (HWHM) [Fig. 4(c)] with the low-energy side of the absorption spectrum extending even further than 750 nm. This result is too broad compared to the Gaussian decomposition results of the experimental linear spectrum in Fig. 1(b), where the Q_x linewidth is estimated to be $\sim 780 \text{ cm}^{-1}$ (HWHM). Therefore, $\tau_{x \rightarrow y}$ has a lower limit below which an unrealistic linear spectrum having the low-energy wing of the Q_x band extends out to the region $\lambda > 720 \text{ nm}$, where there is no absorption experimentally observed (see Fig. 1).

We also numerically simulated for $\tau_{x \rightarrow y} \approx 5$ fs, which yields a realistic linear absorption spectrum comparable to the experimental spectrum [Fig. 4(b)]. However, with a correlation parameter ρ kept as 0.99, the simulated quasi-TA spectra in this case still exhibit λ_{exc} -dependence [Fig. 4(e)], which does not resemble the experimental data. In order to retrieve a similar λ_{exc} -independence as observed in the experimental spectra, the correlation parameter ρ needs to be adjusted to ≤ 0.38 to achieve λ_{exc} -independent quasi-TA spectra resembling Fig. 4(f). These simulations imply that to explain the experimental data of both the linear absorption and quasi-TA spectra, we must assume a low correlation regime of ρ .

DISCUSSION

Within the calculations performed in the highly correlated regime $\rho = 0.99$, to reproduce the similar margin of λ_{exc} -independence between three simulated quasi-TA spectra, the internal conversion rate has to be extremely fast with the time-constant $\tau_{x \rightarrow y}$ of at least 2 fs or below (see Fig. 4) to give results similar to the experimental observations shown in Fig. 2. Even relaxing the time-constant $\tau_{x \rightarrow y}$ to 5 fs to obtain a more realistic spectrum still requires a relatively uncorrelated regime of $\rho = 0.38$. To account for this partial noncorrelation, we first consider if it can be attributed to intramolecular spectral diffusion in the Q_y band after the $Q_x \rightarrow Q_y$ relaxation. A quantitative measure of spectral diffusion is the normalized frequency-fluctuation correlation function (FFCF).^{38,41,42} The normalized FFCF can be interpreted as a T_w -dependent dynamic correlation ρ . For Chl *a* in various solvent conditions, this dynamic correlation has been measured to be $\rho \sim 0.5$ at $T_w = 100$ fs.^{35,43} Hence, at $T_w = 60$ fs, ρ due to spectral diffusion should be even higher than 0.5. This indicates that spectral diffusion can be a contributing factor of the relatively uncorrelated regime, but there likely needs to be another factor to further bring the correlation factor down to $\rho = 0.38$ for the case of $\tau_{x \rightarrow y}$ to 5 fs, which will be discussed below.

However, we deem such scenarios mentioned above, with $\tau_{x \rightarrow y} \sim 5$ fs or shorter, as unlikely, since the rates seem too fast compared to the several spectroscopic measurements of ~ 100 to 200 fs that have been reported for molecular Chl *a*.^{29,30} Although the protein scaffold of the PSII CC can induce some variations of the dielectric environment in the binding pockets and hence influences the intramolecular Chl *a* dynamics, more than an order of magnitude increase of the internal conversion rate compared to dissolved molecular Chl *a* may not be very realistic. On the other hand, by keeping the value of $\tau_{x \rightarrow y}$ to be 20 fs to ensure the complete relaxation at 60 fs delay time as observed in the experimental 2D and quasi-TA spectra, we can reasonably reproduce the λ_T -independent quasi-TA spectra with a low ($<15\%$) correlation between the Q_x and Q_y transitions [see Fig. 3(c)]. Such low $\rho \sim 0.15$ at $T_w = 60$ fs cannot solely be due to spectral diffusion in the Q_y band as already explained above. Based on the simulation results, the λ_T -independent quasi-TA spectra retrieved from the 2D spectrum at $T_w = 60$ fs of PSII CC connecting the Q_x and Q_y bands are explained mainly by the lack of correlation between the two involved transition bands.

We now discuss two possible sources of this lack of correlation. The Chl *a* pigments in a PSII CC exhibit a distribution in frequency space of their Q_y transition that gives rise to the EET network. Each Chl *a*'s Q_y transition will have a concomitant Q_x manifold and an associated difference in energy ΔE_{xy} . We can expect that there will be a distribution in this interband energy ΔE_{xy} between different excitons due to differences in the local environment and differences in couplings between various Chl *a* pigments. This distribution in interband energy provides the first possible source of the lack of correlation. The second source can be attributed locally to an intra-molecule property. Each of the molecule's Q_x and Q_y bands will have a linewidth associated with it. As an example, one can see this in the spectrum of dispersed molecular Chl *a* in THF [Fig. 1(b)]. A previous 2DES study that measured the time-evolution 2D peakshape connecting the Q_x and Q_y bands of dissolved Chl *a* determined that upon the transfer from Q_x to Q_y state, only $\sim 10\%$ to 20% of the correlation remains.³³

For the first source attributed to the distribution in this interband energy ΔE_{xy} , further theoretical calculations and simulations may need to be performed. Unfortunately, such studies are lacking in the literature, as most calculations only report the Q_y band energies.^{3,44,45} For the second source attributed to intrinsic intramolecule correlation loss, we note that the $\rho \approx 0.15$ value in the calculation section above approximately matches the reported $\sim 10\%$ to 20% remaining correlation of molecular Chl *a*.³³ Therefore, if the correlation dynamics for the Chls *a* embedded within the native protein scaffold of PSII CC are similar to those of molecular Chl *a*, the observed lack of correlation is likely mostly contributed by this intrinsic low correlation between the Q_x and Q_y transitions within each Chl *a*. However, a definitive study will only be possible with the availability of the calculations to yield the distribution of this interband energy ΔE_{xy} of PSII CC.

Our current study of the low correlation between Q_x and Q_y bands of the PSII CC might open an interesting avenue to better understand the energy distribution upon excitation of the light-harvesting complexes. Within the Q_y band, which is usually inhomogeneously broadened by various Q_y transitions coming from different pigments, different wavelengths of light can excite the complex differently with a varied combination of localized excited

pigments. The excitons then need time to migrate around the whole complex with the rate governed by spectral overlaps and excitonic contributions^{46–48} to finally reach the reaction center to perform a redox chemical reaction in the case of PSII CC. A time-resolved fluorescence study determined a significant slowing down of the exciton migration rate with an increase of the antenna size of PSII-LHCII supercomplexes.⁴⁹ This poses potential ineffective energy migration pathways in non-optimal light conditions. On the other hand, the excitations in the Q_x band are generally non-selective, with almost the whole (set of pigments) complex being excited. Once the excitation quickly relaxed into the lowest electronic potential according to Kasha's rule (Q_y in the Chl *a* case),⁵⁰ all pigments have an equal chance (weighted by the transition dipole moments) to be excited regardless of the light conditions. This implies that at least ~20% of absorbed light (as proportion absorbed by the Q_x compared to Q_y band) are distributed throughout the whole complex's energy manifold and can avoid any "bottleneck" in the photosynthetic complexes originated from inefficient excitonic couplings between the Q_y transition dipole moments. Although in natural systems, such as PSII CC, the transition dipole moments of Q_x bands are relatively low, this should not be a problem for artificial chemical-engineered light-harvesting devices and the low correlation between different transitions still holds the value of uniform excitation ability.

CONCLUSION

In this study, we report the 2DES measurement of the monomeric PSII CC at physiological temperature. With the high time-resolution due to the ultra-broadband light source, the 2D spectra reveal an ultrafast relaxation (<60 fs) from the high-energy Q band, which is comprised of Q_x and vibrational progressions of Q_y transition, to the main Q_y absorption band.

Detailed spectral analysis reveals that the ultrafast internal conversion of PSII CC Chl *a* Q_x to Q_y is not frequency selective. Using numerical modeling with varying correlation between Q_x and Q_y transitions, as well as the $Q_x \rightarrow Q_y$ relaxation rate, we determine that the lack of frequency selectivity probably originated from the lack of correlation between Q_x and Q_y transitions instead of the fast relaxation induced broadening of the homogeneous spectral linewidth. In general, the excitation-independent quasi-TA spectra obtained from Q_x excitation can be reproduced with a low correlation parameter of below 15% between Q_x and Q_y transitions, given a realistic $Q_x \rightarrow Q_y$ relaxation timescale of ~20 fs. It is likely that this loss of correlation is partly due to the intrinsic intramolecular low correlation between Q_x and Q_y transitions that was previously reported. In that study, ~80% to 90% correlation loss was observed upon ultrafast internal conversion between Q_x and Q_y bands of molecular Chl *a* in organic solvents.³³ By virtue of the fact that there are only Chls *a* in PSII CC, it allows us to clearly observe the lack of frequency selectivity of the Chl *a* $Q_x \rightarrow Q_y$ relaxation without interference from the Chl *b* absorption band.

In general, we would similarly expect this lack of frequency selectivity in other biological pigment-protein complexes, such as the PSII's antenna complexes (LHCII, CP29, CP24, and CP26) and PSI supercomplex, which have similar protein-embedded Chls *a* pigments. The ultrafast non-selective $Q_x \rightarrow Q_y$ internal conversion can enhance the EET efficiency of the photosynthetic complexes by ensuring that a significant amount of absorbed light can be

distributed on the whole complex without going through some inefficient EET pathways. This may imply a change in how we understand the overall EET dynamics of the light-harvesting complexes, which is hitherto extensively studied assuming only Q_y dynamics.^{11,12,47}

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

T.N.D. and H.L.N. contributed equally to this work.

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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